

SungJin Noh

노성진

M.S. student, Computational Robotics and Autonomy Lab
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EDUCATION

M.S., Electrical and Electronic Engineering Mar. 2025 – Feb. 2027 (expected)
Inha University, Incheon, Korea. Advisor: Prof. Kwang-Ki K. Kim.

B.S., Electrical Engineering Mar. 2019 – Feb. 2025
Inha University, Incheon, Korea.

PUBLICATIONS

Journal

- [J1] **SungJin Noh**, JunGee Hong, and Kwang-Ki K. Kim, “Attitude-aware Second-order Conic-constrained Trajectory Optimization for 6-DOF Quadrotor Precision Landing,” *Journal of Korea Robotics Society*, vol. 21, no. 1, pp. 100–108, 2026. doi:10.7746/jkros.2026.21.1.100
Presented at the 21st Korea Robotics Society Annual Conference (KRoC), Pyeongchang, Feb. 2026, under the journal’s simultaneous submission option. Korean patent application pending, Ref. KR07990 / PD26-154-N.

Conference

- [C1] **SungJin Noh**, Yong-Hee Kim, Yeohosua Kim, and Kwang-Ki K. Kim, “Compound State-Triggered Constrained DDP for Receding-Horizon Quadrotor Landing on a Moving Platform,” *26th International Conference on Control, Automation and Systems (ICCAS)*, Sapporo, Japan, Oct. 2026. (accepted)

RESEARCH EXPERIENCE

Computational Robotics and Autonomy Lab, Inha University Mar. 2025 – present

- **State-triggered constraints in DDP.** Formulated the final-approach requirement as a compound state-triggered constraint whose violation is integrated as a continuous-time accumulator, so a glideslope cone binds only inside the approach region and the constraint stays stage-separable through the DDP backward pass. Implemented it in an existing interior-point DDP solver and ran the receding-horizon evaluation over six target motions, grading landing and constraint satisfaction together. First author, ICCAS 2026.
- **Second-order cone constrained precision landing.** Formulated the landing problem with landing-cone and thrust-tilt constraints kept in exact SOC form over quaternion attitude dynamics, with planning decoupled from control allocation, and implemented it in an interior-point DDP solver. Ran the simulation study and the hardware experiments on a Crazyflie 2.1 with Lighthouse positioning, including recovery from large initial tilt. First author, *J. Korea Robotics Society* 2026; patent pending.
- **ROS integration.** Built ROS 2 and ROS 1 wrappers that run the CoManDO planner on Crazyflie and PX4, keeping the command stream running while a solve is in flight. Exercised in CrazySim software-in-the-loop and on hardware with Qualisys motion capture.

TEACHING

Teaching assistant (grading), Inha University: Robotics, Spring 2026; Numerical Analysis (EEEC3600), Spring 2025 and Spring 2026.

SKILLS

- **Programming.** C++, Python, MATLAB, Julia, $\text{\LaTeX}$
- **Optimization and control.** Differential dynamic programming and iLQR, interior-point methods, augmented Lagrangian methods, successive convexification, model predictive control, second-order cone programming
- **Software.** ALIPDDP, acados, OCS2, CVXPY, IPOPT; ROS 2 / ROS 1, PX4, MAVROS, Gazebo, Crazyswarm and CrazySim
- **Languages.** Korean (native), English (professional working proficiency)